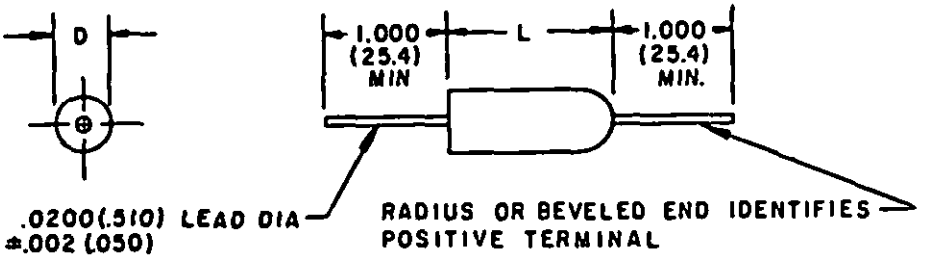
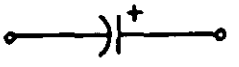
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, MOLDED, NONHERMETICALLY SEALED, STYLE CX0S

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-49137.





CIRCUIT DIAGRAM

Case size	Dimensions	
	Diameter D ±.015 (.38)	Length L ±.020 (.51)
A	.085 (2.16)	.250 (6.35)
B	.100 (2.54)	.280 (7.11)
C	.170 (4.32)	.335 (8.51)
D	.170 (4.32)	.410 (10.41)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

TABLE I. Style CX05 characteristics.

Type designation 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max) +25°C	Dissipation factor	Case size
	Volts	µF	µA	%	
CX05D475-	6	4.7	1.0	6	A
CX05D156-	6	15	1.0	8	B
CX05D336-	6	33	1.5	8	C
CX05D476-	6	47	3.0	8	D
CX05F335-	10	3.3	1.0	6	A
CX05F106-	10	10	1.0	8	B
CX05F226-	10	22	1.5	8	C
CX05F336-	10	33	3.0	8	D
CX05H225-	15	2.2	1.0	6	A
CX05H685-	15	6.8	1.0	6	B
CX05H156-	15	15	1.5	8	C
CX05H226-	15	22	3.0	8	D
CX05J155-	20	1.5	1.0	6	A
CX05J475-	20	4.7	1.0	6	B
CX05J156-	20	15	3.0	8	D
CX05K105-	25	1.0	1.0	4	A
CX05K225-	25	2.2	1.0	6	B
CX05K335-	25	3.3	1.0	6	B
CX05K685-	25	6.8	1.5	6	C
CX05K106-	25	10	1.5	8	C
CX05M334-	35	0.33	1.0	4	A
CX05M474-	35	0.47	1.0	4	A
CX05M155-	35	1.5	1.0	6	B
CX05M335-	35	3.3	1.5	6	C
CX05M475-	35	4.7	1.5	6	C
CX05M685-	35	6.8	3.0	6	D
CX05N104-	50	0.10	1.0	4	A
CX05N154-	50	0.15	1.0	4	A
CX05N224-	50	0.22	1.0	4	A
CX05N334-	50	0.33	1.0	4	B
CX05N474-	50	0.47	1.0	4	B
CX05N684-	50	0.68	1.0	4	B
CX05N105-	50	1.0	1.0	4	B
CX05N155-	50	1.5	1.5	6	C
CX05N225-	50	2.2	1.5	6	C
CX05N335-	50	3.3	2.0	6	D
CX05N475-	50	4.7	3.0	6	D

1/ Complete type designation will include an additional symbol indicating capacitance tolerance.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Case material: Molded, epoxy or plastic.
Leads: Solder-coated metal.
DC rated voltage: See table I.
Operating temperature: -55° to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.
Cap. tolerance: ±10% (K), or ±20% (M).

Dissipation factor (DF): See table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

Stability at low and high temperature: In accordance with MIL-C-49137.

Step 1 (+25°C):	DCL	- See table I.
	Cap.	- Within tolerance specified in table I.
	DF	- See table I.
Step 2 (-55°C):	ΔCap.	- Within -15 percent of step 1 measured value.
	DF	- 200 percent of table I.
Step 3 (+25°C):	DCL	- See table I.
	ΔCap.	- Within 10 percent of step 1 measured value.
	DF	- See table I.
Step 4 (+85°C):	DCL	- 10 times 25°C limit (see table I).
	ΔCap.	- Within +15 percent of step 1 measured value.
	DF	- 150 percent of table I.
Step 5 (+25°C):	DCL	- See table I.
	ΔCap.	- Within 10 percent of step 1 measured value.
	DF	- See table I.

Surge voltage: In accordance with MIL-C-49137.

DCL	- See table I.
ΔCap.	- Within 10 percent of initial value.
DF	- See table I.

Life: Method 108 of MIL-STD-202.

1,000 hours:

At 25°C:	DCL	- 200 percent of table I.
	ΔCap.	- Within 10 percent of initial measured value.
	DF	- See table I.

NOTES:

1. Revision letters are not used to denote changes due to the extensiveness of the changes.
2. These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - ER
Navy - SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Army -
Navy - MC, AS, OS, CG
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-1257-6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-49137/5 CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, MOLDED, NONHERMETICALLY SEALED, STYLE CX05

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Cdr, US Army ERADCOM
ATTN: DELET-RP
Fort Monmouth, NJ 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:50Z
Check the source to verify that this is the current version before use.